# P001-MVB

Responsible AI Clinical Decision Support Prototype

## Explainable and Responsible AI for Differentiating Bacterial and Viral Meningitis: A Clinical Decision Support System


Presenter

Abubakar Amidu

Mentor

Chukwukadibia Onyekwere


Programme

3MTT DeepTech Cohort 2 – DS/ML Mentorship

Location

Abuja, Nigeria

Accurate Prediction

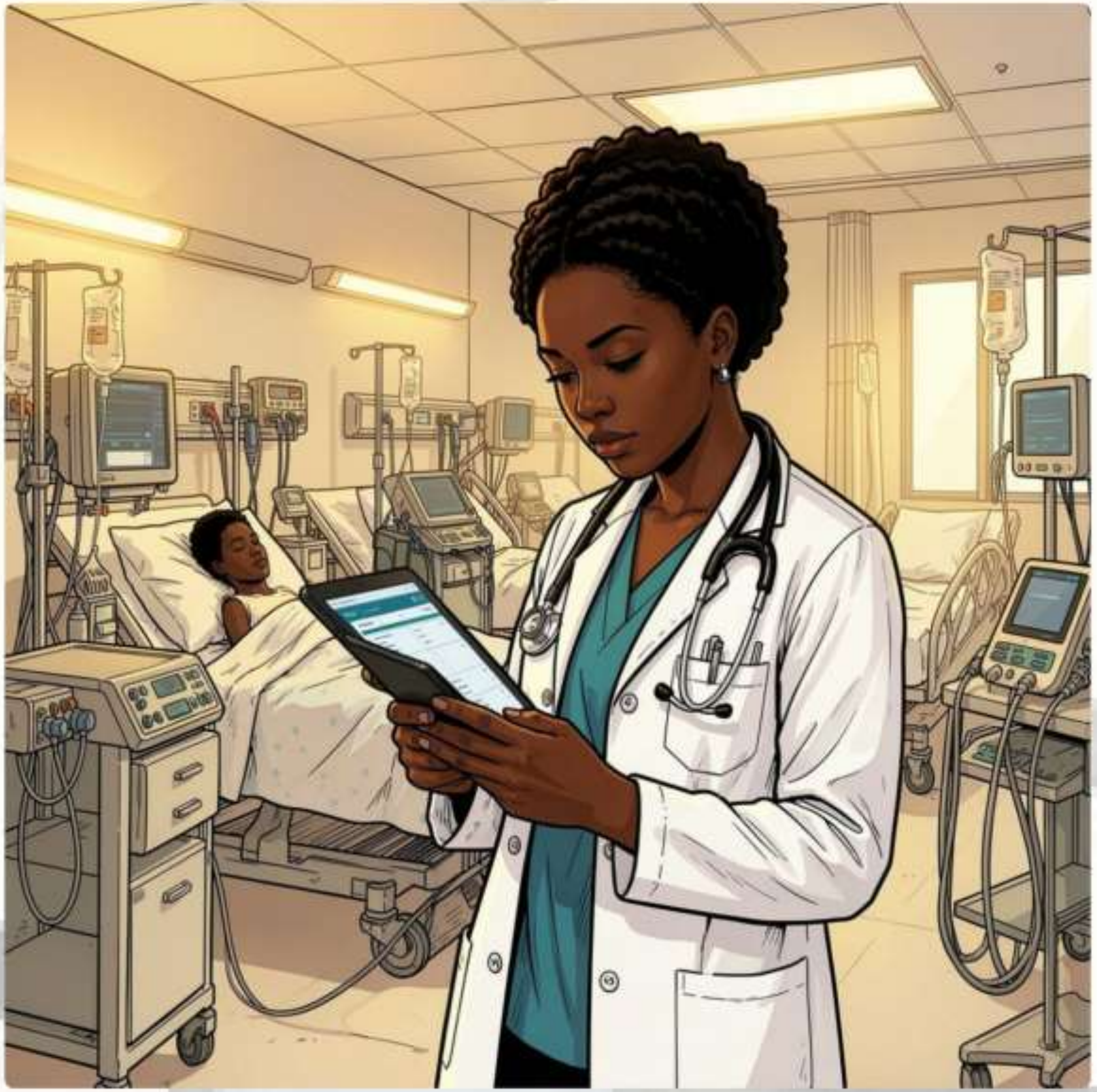
Explainable AI

Responsible Governance

Human Oversight

# The Problem

- ⚠ **Delayed Diagnosis** — Rapidly distinguishing **bacterial** from **viral meningitis** is clinically challenging, yet treatment decisions must often be made **urgently**.
- ⚠ **Black-Box AI** — Many healthcare AI systems operate as **black boxes**, offering little explanation for their predictions.
- ⚠ **Limited Transparency** — This lack of transparency reduces clinician trust and confidence, increases automation bias, and limits safe adoption in healthcare.



# Our Solution — P001-MVB

P001-MVB integrates four complementary components delivered through an interactive **Streamlit** clinical decision-support application, prioritising transparency, explainability, governance, and mandatory human oversight.

### Explainable Machine Learning

Logistic Regression with interpretable outputs

### SHAP Explanations

Feature-level transparency for every prediction

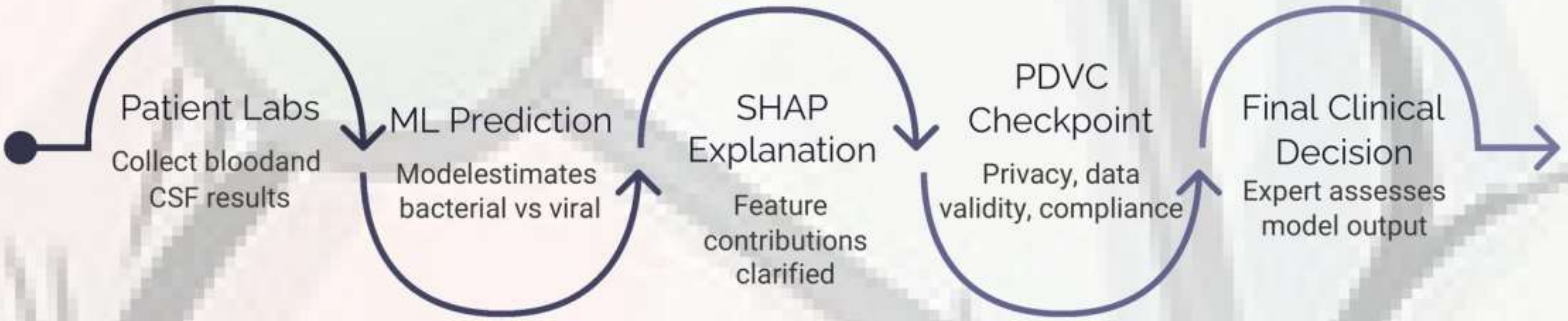
### Responsible AI Governance

Model Card, AI Risk Assessment, structured governance

### PDVC

Pre-Decision Verification Checkpoint — mandatory clinician review

# P001-MVB Clinical Decision Workflow




Every prediction passes through explainability and a mandatory verification checkpoint before influencing any clinical decision — supporting, not replacing, clinical judgement.

# Tools & Technologies

## Programming & ML

Python · Pandas · NumPy · Scikit-learn · XGBoost · SHAP

## Deployment

Streamlit Community Cloud · GitHub

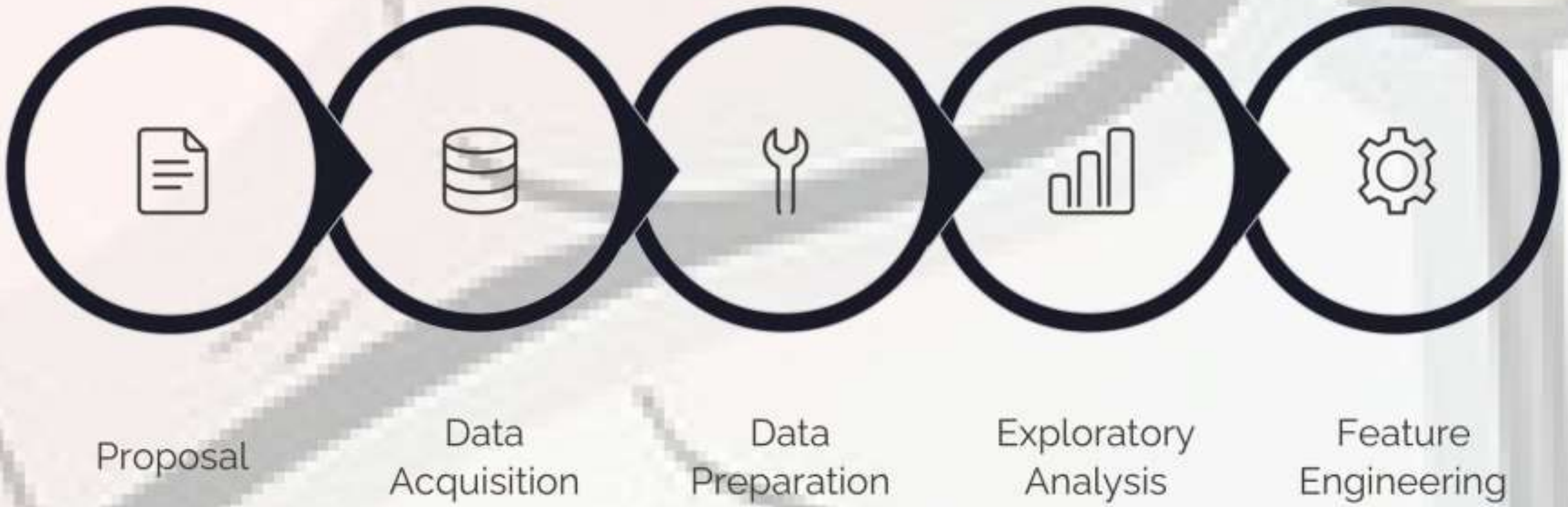
## Development

Google Colab · Jupyter Notebook

## Responsible AI

Model Card · AI Risk Assessment · PDVC Framework

# End-to-End AI Development Lifecycle




Six machine learning models were evaluated. The final model was selected based on **Bacterial Recall** — prioritising patient safety over overall accuracy.

### Near-Proxy Removal

Removed **Pathogen_Present** feature to prevent data leakage

### Feature Set Comparison

Full vs Restricted feature sets rigorously evaluated

### Safety-First Model Selection

Bacterial Recall prioritised — minimising missed diagnoses

### **Responsible AI Governance**

Embedded governance integrated directly into the deployed application

## Dataset

Source: Kaggle – Meningitis Classification (ChanTest)

Original records: 1,200

Removed: 67 (“Unknown” diagnosis)

Final records: 1,133

Bacterial: 595 · Viral: 538

# Results & Live Application Demonstration

Final Selected Model: Logistic Regression (Restricted Feature Set)

Streamlit Home

Prediction Result

SHAP Waterfall

PDVC Verification

Pre-Decision Verification Checkpoint (PDVC) automatically requiring clinician review when governance triggers are activated, with an audit log supporting transparency and accountability.

**93.4%**
Accuracy

Overall model accuracy on held-out test data

**95.8%**
Bacterial Recall

Critical safety metric – minimising missed bacterial cases

**91.9%**
Precision

Proportion of bacterial predictions that are correct

**5**
PDVC Triggers

Automated governance triggers plus clinician-initiated escalation

### Responsible AI Outcomes

- SHAP provides transparent **feature-level explanations** for every prediction
- All predictions include **confidence scores**
- All five false-negative evaluation cases would have **triggered PDVC verification** before influencing clinical decisions

All five false-negative cases would have been flagged by the PDVC – demonstrating the safety net in practice.

**Live Application:** Screenshots of the Streamlit Home Screen, Prediction Results, SHAP Explanation, and PDVC Verification screens are available in the live demonstration.

## Models Evaluated

Logistic Regression – Full / Restricted

Random Forest – Full / Restricted

XGBoost – Full / Restricted

**Selected: Logistic Regression (Restricted) — best balance of bacterial recall, interpretability, transparency and Responsible AI deployment requirements.**

# Pre-Decision Verification Checkpoint (PDVC)

Manyhealthcare AIprojectsstopafterproducinganaccurateprediction.**P001-MVB**goesfurtherbyintegratingResponsible AI governance directly into the clinical decision-support workflow.

### Detects

Higher-risk predictions automatically

### Requires

Mandatory clinician verification before action

### Promotes

Transparency through embedded explainability

### Reduces

Automation bias and over-reliance on AI

### Reinforces

Clinician authority and accountability

> **Governance is not documentation—it is an operational safeguard embedded in the deployed AI system.**

# Conclusion & Future Work

→ Key Takeaway 1

Explainable AI improves trust by making each prediction transparent to clinicians.

→ Key Takeaway 2

Safety-first model selection prioritises bacterial recall to reduce missed critical cases.

→ Key Takeaway 3

Responsible AI governance is embedded directly into the workflow through PDVC.

→ Key Takeaway 4

Human oversight remains central, supporting clinical judgement rather than replacing it.

01

## Expand Dataset Coverage

Collect additional clinical cases to improve robustness and generalisability.

02

## Validate Across Sites

Test performance with data from multiple hospitals and care settings.

03

## Refine Governance Rules

Enhance PDVC triggers, thresholds, and escalation logic for safer deployment.

04

## Improve Model Performance

Compare additional interpretable models and calibration strategies.

05

## Prepare Clinical Integration

Develop workflow-ready deployment features for real-world use.

### Limitations

Research prototype only — not intended for clinical use. · Dataset is clinically inspired rather than clinically validated.
No external validation has yet been performed. · Future work includes validation using representative clinical datasets.

# Thank You

## Questions?

**Abubakar Amidu** · 3MTT DeepTech Cohort 2 – DS/ML Mentorship · Abuja, Nigeria

# Live Resources

GitHub Repository

https://github.com/Abamid/P001-MVB

Live Streamlit Application

https://p001-mvb-ig6zigmh6yr9tbdamggagc.streamlit.app/

**Scan the QR codes to explore the complete project.**